

# Tyler Han

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## EDUCATION

### • University of Washington

Ph.D. Computer Science

June 2022 - Present

Advisor: [Byron Boots](#)

### • University of Maryland

B.S. Aerospace Engineering | B.S. Computer Science | Minor: Mathematics

Sep 2018 - May 2022

GPA: 3.9 / 4.0

## RESEARCH INTEREST

I am interested in methods which enable robots to attain the same level of efficiency and robustness as animals. Supported by my background in aerospace engineering and field robotics, I am strongly motivated by real-world deployment.

## PUBLICATIONS

1. Tyler Han, Yanda Bao, Bhaumik Mehta, Gabriel Guo, Anubhav Vishwakarma, Emily Kang, Sanghun Jung, Rosario Scalise, Bryan Xu, Byron Boots. "Model Predictive Adversarial Imitation Learning for Planning from Observation", *International Conference on Learning Representations (ICLR)*, 2026. [\[pdf\]](#)
2. Tyler Han, Preet Shah, Sidharth Rajagopal, Yanda Bao, Sanghun Jung, Sidharth Talia, Gabriel Guo, Bryan Xu, Bhaumik Mehta, Emma Romig, Rosario Scalise, Byron Boots. "Wheeled Lab: Modern Sim2Real for Low-Cost, Open-source Wheeled Robotics", *Conference on Robot Learning (CoRL)*, 2025. [\[pdf\]](#) [\[website\]](#)
3. Tyler Han, Alex Liu, Alex Spitzer, Guanya Shi, Byron Boots. "Model Predictive Control for Aggressive Driving Over Uneven Terrain", *Robotics: Science and Systems (RSS)*, 2024. [\[pdf\]](#) [\[website\]](#)
4. Chuning Zhu, Xinqi Wang, Tyler Han, Simon Du, Abhishek Gupta. "Distributional Successor Features Enable Zero-Shot Policy Optimization", *Neural Information Processing Systems (NeurIPS)*, 2024. [\[pdf\]](#) [\[website\]](#)

## PREPRINTS

1. Tyler Han, Bat Nemekhbold, Siyang Shen, Rohan Baijal, Richard Ebock, Harine Ravichandiran, Sanghun Jung, Kevin Huang, Byron Boots. "Online World Modeling Enables Real-World Inverse Reinforcement Learning from Observation", 2026. [\[website\]](#)
2. Anubhav Vishwakarma, Caleb Hsu, Byron Boots, Karen Leung, Tyler Han. "Discriminative Barrier Functions for Safe Adversarial Imitation Learning from Observation", 2026. [\[pdf\]](#)
3. Tyler Han, Sidharth Talia, Rohan Panicker, Preet Shah, Neel Jawale, Byron Boots. "Dynamics Models in the Aggressive Off-Road Driving Regime", *International Conference on Robotics and Automation (ICRA). Workshop on Off-Road Autonomy*, 2024. [\[pdf\]](#)

## HONORS AND AWARDS

### • Best Paper Finalist

Resource-Rational Robot Learning, CoRL 2025

September 2025

### • Graduate Research Fellowship Program (GRFP)

National Science Foundation

March 2023

## SERVICE

- Guest Speaker at Global Innovation Exchange, University of Washington
- Organizer and Award Committee Member: RSS 2025 Off-Road Autonomy Workshop
- Invited to Review: RA-L, T-RO, ICRA, RSS Off-Road Workshop, CoRL, NeurIPS

## MEDIA

- "National Robotics Week – Latest Physical AI Research, Breakthroughs and Resources", NVIDIA, 2025 [\[link\]](#)
- "From Simulation to Reality: Building Wheeled Robots with Isaac Lab", Q&A Talk, 2025 [\[link\]](#)

## OUTREACH & MENTORING

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- **Robotics Research Mentoring**

June 2023 - Present

*Robot Learning Lab*

- Master's Students: Siyang Shen → Paxini, Anubhav Vishwakarma → FieldAI, Preet Shah → Research Engineer at General Robotics, Alex Liu → Software Engineer at Amazon
- Undergraduate Students: Laura Wang, Urvi Rutia, Bhaumik Mehta, Bryan Xu, Yanda Bao, Sidharth Rajagopal, Gabe Guo, Alyssa Giedd → Ph.D. at University of Washington
- High School Students: Vansh Chhabra

- **Pre-Application Mentorship Program**

September 2022 - Present

*University of Washington, Department of Computer Science & Engineering*



- Mentor to students in under-represented or underserved communities considering graduate school

## PROJECTS

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- **Wheeled Lab**

February 2025 - Present

*University of Washington*



- Created education platform for students new to Reinforcement Learning in robotics
- Mentor undergraduate teams beginning in robotics (not included in above "Outreach & Mentoring" section) through Wheeled Lab by advising extracurricular projects in Sim2Real

- **Robotic Autonomy in Complex Environments with Resiliency (RACER)**

June 2022 - Present

*Defense Advanced Research Projects Agency (DARPA)*



- Algorithmic and software development of control system for state-of-the-art unmanned ground vehicle (UGV)
- Conduction of day and week long field tests for controls system and perception & planning integration

- **Autonomous Tunnel Boring**

Sep 2020 - May 2022

*The Boring Company | Not-a-Boring Competition*

- Development of globalization, dynamics, and control system for autonomous tunnel boring (1 m diameter)

- **Deep Motor Primitives**

Aug 2020 - Dec 2021

*Naval Research Laboratory*

- Develop algorithms for learning motor primitives using Dynamic Mode Decomposition and its derivatives

- **Manipulation for Satellite Servicing**

Jan 2019 - Aug 2020

*UMD Space Systems Laboratory*

- Infrastructural and algorithmic software development for 8-DOF dexterous satellite servicing manipulator